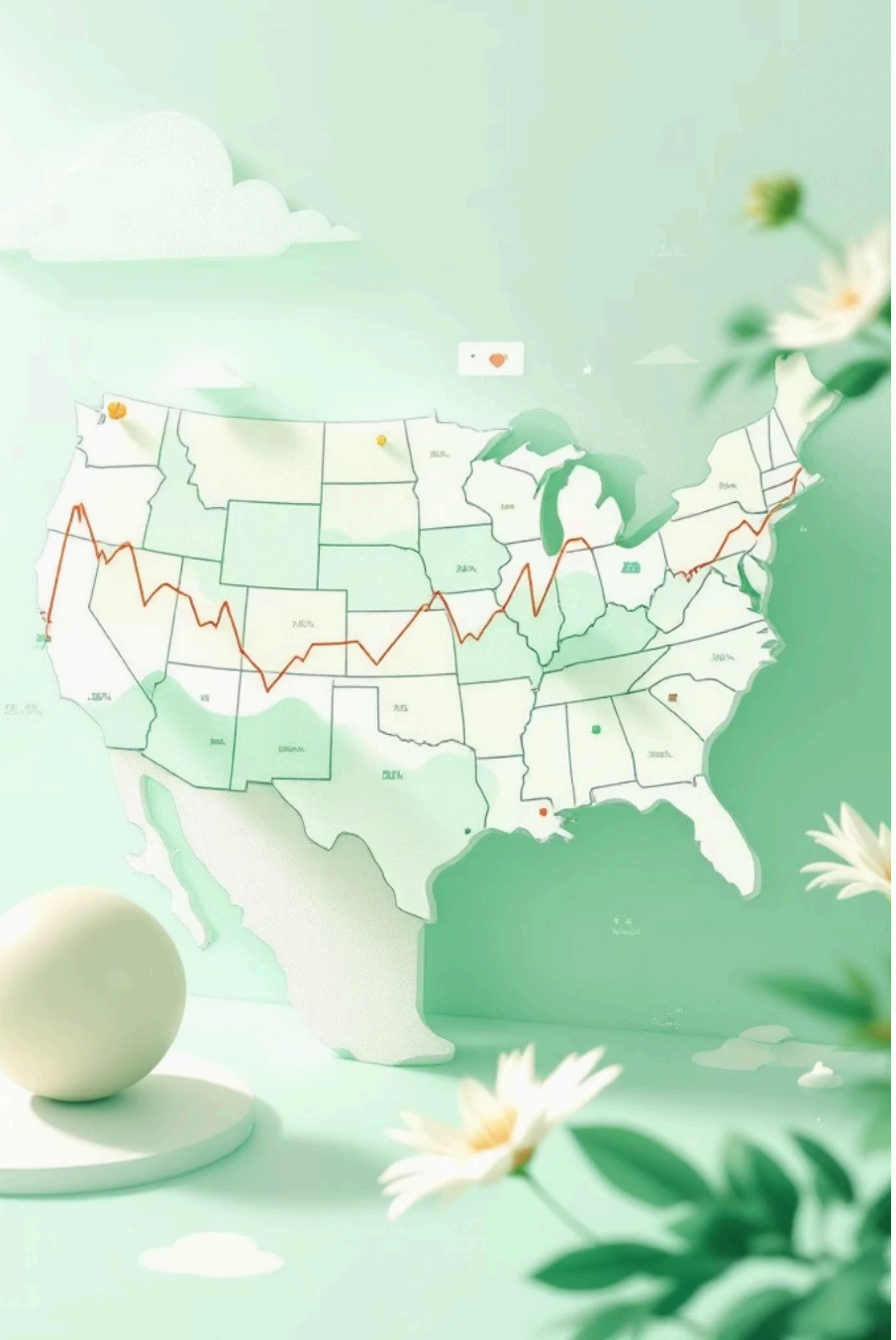

Has Inflation Permanently Changed Consumer Behaviour in the U.S.?

A Counterfactual Analysis Using FRED Macroeconomic Data

Spendemic | March 3rd 2026

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A Crisis That Raised a Lasting Question

The COVID-19 pandemic triggered the fastest inflation surge in 40 years. Prices rose. Rates spiked. Stimulus flooded households. But when the dust settled – did Americans go back to normal, or did something permanently change?

Spending

Saving

Borrowing

Pre-COVID

Stable growth and low inflation

COVID Shock

Rapid disruption and uncertainty

Inflation Surge

Prices surged and rates rose

Post-Inflation

Transition toward normalization

Understanding the lasting impact of unprecedented economic events requires rigorous analysis.

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12 FRED Indicators. 3 Eras. One Question.

Datasets Used

- Real Personal Consumption Expenditures (PCE)
- Personal Saving Rate
- Consumer Credit Outstanding
- CPI (Overall, Food, Energy, Shelter)
- Federal Funds Rate
- Real Disposable Income
- Unemployment Rate
- CC Delinquency Rate
- Consumer Sentiment Index (UMich)
- Retail & Food Sales
- Average Hourly Earnings

Methodology Stack

Step 1: EDA + Era Segmentation

2010 - 2024, defining three distinct economic eras.

Step 2: Linear Regression Counterfactual Baseline

Establishing a 'no-shock' trend for comparison.

Step 3: Auto-SARIMAX Counterfactual Modeling

Incorporating exogenous variables for refined predictions.

Era	Period
Pre-COVID	Jan 2010 - Feb 2020
COVID/Inflation Shock	Mar 2020 - Jun 2023
Post-Shock	Jul 2023 - Present

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The Shock Was Real, Large, and Persistent

CPI Rose +32.8% Overall

Pre-COVID avg: 237.5 → Post-Shock avg: 315.4

Fed Funds Rate Up +669%

Pre-COVID avg: 0.63% → Post-Shock avg: 4.84% | Peak: 5.33%

Real Disposable Income Surged Then Stalled

Stimulus spike: +30.7% in 13 months → Post-shock growth: only +5.5%

Key Insight

Prices settled at a permanently higher level. The era of cheap money is over. Stimulus income was temporary – but the behavioral response was not.

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Spending Recovered — But Deviated From Its Historical Path

$12,543B

Pre-COVID Avg PCE

$16,171B

Post-Shock Avg PCE

(**+28.9%** overall)

+9.96%

Post-Shock Growth Rate

(Slower than pre-shock)

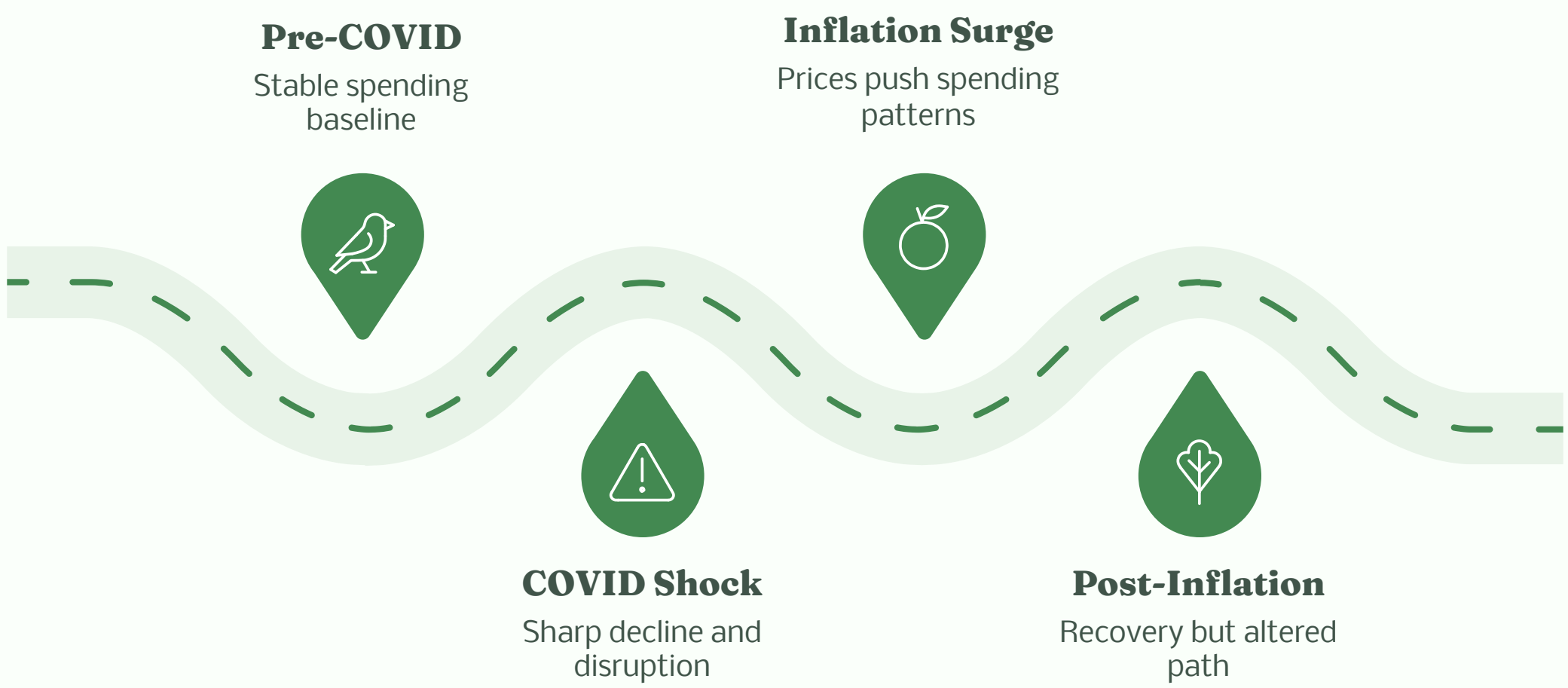
Model Result: Consumer Spending

- **Linear Regression Counterfactual:** R^2 = 0.018 post-COVID (pre-COVID trend explains only 1.8% of variance)
- RMSE = 981 | MAE = 898
- **SARIMAX Gap Analysis:**
- COVID period: -4.13% below trend
- Post-Shock: -5.87% below trend (widening)
- Latest gap: -6.28%

GAP WIDENING → STRUCTURAL SHIFT

Key Insight

Nominal spending looks strong, but it has structurally deviated from its pre-COVID trajectory. Real purchasing power gains are minimal once inflation is factored in.



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Americans Are Saving Less Than Ever Before

6.16%

Pre-COVID Avg

31.8%

COVID Peak

(April 2020 – stimulus + lockdowns)

5.11%

Post-Shock Avg

(**-17%** below pre-COVID norm)

3.5%

Current Saving Rate

Lowest level in the entire dataset

Model Result: Saving Rate

- **Linear Regression:** R^2 = -0.078 (model performs worse than a flat average. Pre-COVID trend is completely broken.)
- **SARIMAX Gap Analysis:**
- COVID period: +22.59% above trend (stimulus distortion)
- Post-Shock: -41.83% below trend
- Latest gap: -56.20% and still widening

GAP WIDENING → STRUCTURAL SHIFT

Key Insight

The COVID saving surge was artificial – driven by stimulus and forced savings. What followed is a structural decline. Households have less financial buffer than at any point in the past 15 years.

Pre-COVID
Stable savings baseline

COVID Shock
Stimulus-driven peak

Inflation Surge
Rapid decline in savings

Post-Inflation
Lowest saving rates

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Borrowing Hit All-Time Highs — And Kept Climbing

$3.32T

Pre-COVID Avg

$5.02T

Post-Shock Avg

(**+51.2%** overall)

$5.08T

Current Credit

All-time high

Credit growth accelerated even as Fed rates hit 5.33%.

Model Result: Consumer Credit

- **Linear Regression:** R^2 = 0.338 (partial predictability – credit rose faster and higher than projected)
- **SARIMAX Gap Analysis:**
- COVID period: -346% gap (enormous structural above-trend borrowing)
- Post-Shock: -186% gap (persistent, not converging)
- **Fed Funds Rate coefficient:** +49,106 (households borrowed MORE despite higher rates – unusual behavior signaling necessity-driven borrowing)

GAP PERSISTENT → POSSIBLE STRUCTURAL CHANGE

Key Insight

Households are using credit to fill the gap left by depleted savings and inflation-eroded purchasing power. Rising rates haven't slowed borrowing – they've made it more expensive but not less necessary.

Pre-COVID

Steady borrowing growth

COVID Shock

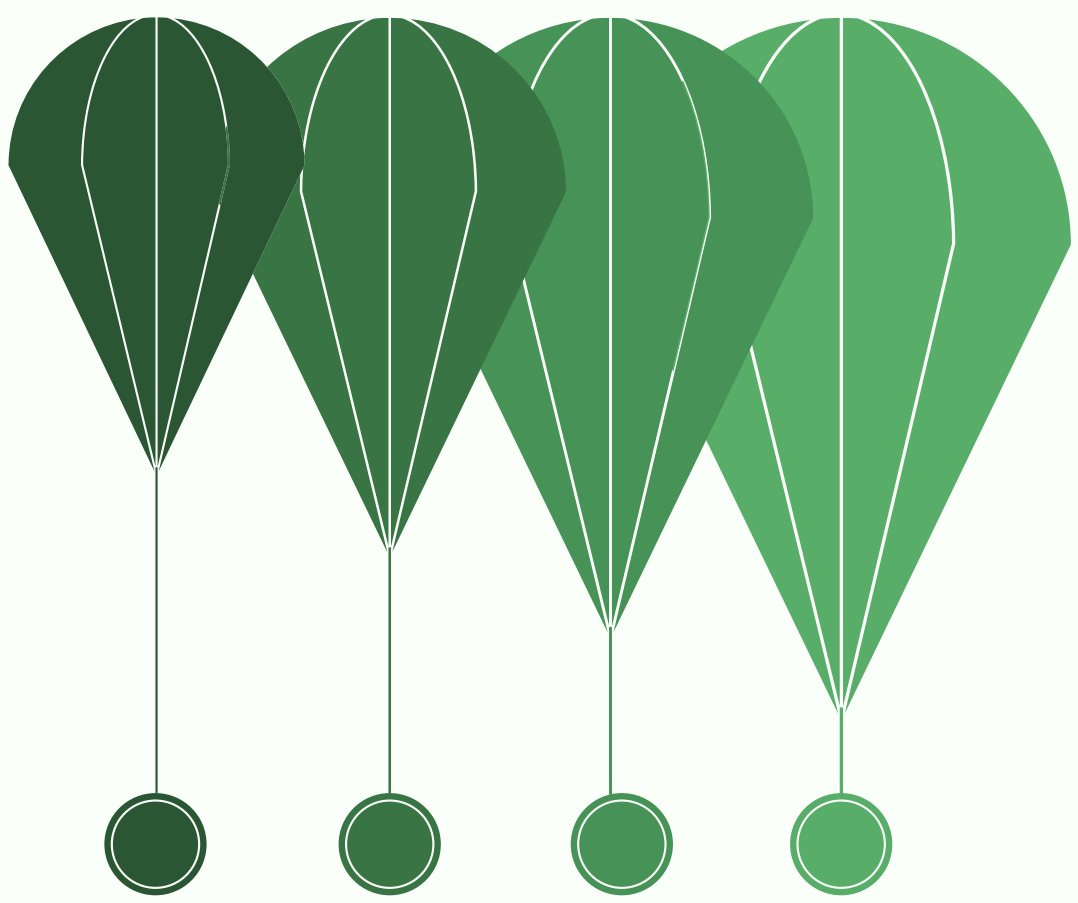
Sharp contraction then rebound

Inflation Surge

Rising costs and strain

Post-Inflation

Recovery and stabilization



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Confidence Is Down. Stress Is Rising.

Consumer Sentiment

Pre-COVID avg: 85.7

Post-Shock avg: 65.9 (**-23.1%**)

Current: 51.0 – approaching pandemic lows

"Americans feel worse about the economy despite low unemployment."

Delinquency Rate

Pre-COVID avg: 2.77%

COVID low: 2.04% (stimulus masked stress)

Post-Shock avg: 3.08% (**+51%** surge from shock era)

"Stimulus relief was temporary. Financial stress is returning."

Unemployment Rate

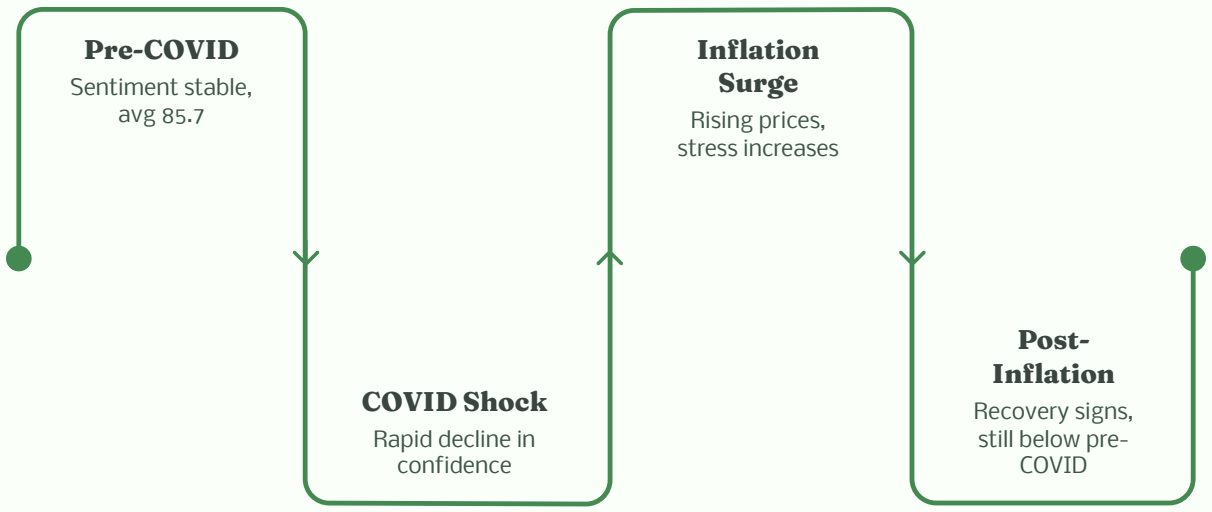
Pre-COVID avg: 6.18%

Post-Shock avg: 4.05% (actually improved)

"Low unemployment hasn't offset the cost-of-living squeeze."

Key Insight

The paradox: jobs are plentiful, but consumer confidence is near recessionary lows. Rising delinquencies confirm that household finances are under real structural strain.



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Interactive Dashboard — Explore the Data

Built with FRED data – interactive era filters, indicator comparisons, and trend breakdowns available in the live dashboard:

https://public.tableau.com/app/profile/soniya.khadka1086/viz/DataJamFeb/Dashboard1

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